Bash Script: Definition and Types

Definition:

A Bash script is a text file containing a series of commands that are executed by the Bash shell to automate tasks or perform operations in Linux or Unix systems.

Structure:

```
#!/bin/bash
# This is a comment
echo "Hello, World!"
```

How to Run:

1. Create a file: nano myscript.sh
2. Add commands inside it.
3. Give permission: chmod +x myscript.sh
4. Run: ./myscript.sh

Types of Bash Scripts

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1. Startup (Initialization) Scripts

- Run automatically when the system or shell starts.

Examples: /etc/profile, ~/.bashrc

2. System Administration Scripts

- Used by admins for automation (backups, updates, etc.).

Example:

```
tar -czf backup_$(date +%F).tar.gz /home/user/Documents
```

3. Automation Scripts

- Automate repetitive user tasks.

Example:

```
for file in *.txt; do mv "$file" processed_"$file"; done
```

4. User Interaction Scripts

- Take user input and give output.

Example:

```
read -p "Enter your name: " name
echo "Welcome, $name!"
```

5. Monitoring or Utility Scripts

- Monitor system resources or logs.

Example:

```
df -h > disk_usage.log
```

6. Network Scripts

- Handle or monitor network tasks.

Example:

```
ping -c 4 google.com > network_check.log
```

7. Conditional and Control Scripts

- Use logic, loops, and conditions.

Example:

```
if [ -f "/etc/passwd" ]; then
  echo "File exists!"
else
  echo "File not found!"
fi
```

Summary Table

Type	Description	Example Use Case

Startup Script	Runs at shell startup	.bashrc
System Admin Script	Automates maintenance	Backups
Automation Script	Repeats common tasks	File renaming
User Interaction Script	Takes input	Setup tools
Monitoring Script	Tracks system status	Disk usage
Network Script	Network monitoring	Ping tests
Control Script	Uses logic & loops	Conditional jobs